Example 7.6.3. 1. Let V be a finite dimensional representation of a group G or a Lie algebra $\mathfrak{g}$. Then the left and right adjoint to the functor $V\otimes$ on the category of representations of G is the functor $V^*\otimes$.

2. The functor Res_K^G is left adjoint to Ind_K^G. This is nothing but the statement of the Frobenius reciprocity.

3. Let $\mathbf{Assoc}_k$ be the category of associative unital algebras, and let $\mathbf{Lie}_k$ be the category of Lie algebras over some field k. We have a functor $L : \mathbf{Assoc}_k \to \mathbf{Lie}_k$ which attaches to an associative algebra the same space regarded as a Lie algebra, with bracket $[a, b] = ab - ba$. Then the functor L has a left adjoint, which is the functor U of taking the universal enveloping algebra of a Lie algebra.

4. We have the functor $GL_1 : \mathbf{Assoc}_k \to \mathbf{Groups}$, given by $A \mapsto GL_1(A) = A^\times$. This functor has a left adjoint, which is the functor $G \mapsto k[G]$, the group algebra of G.

5. The left adjoint to the forgetful functor $\mathbf{Assoc}_k \to \mathbf{Vect}_k$ is the functor of tensor algebra: $V \mapsto TV$. Also, if we denote by $\mathbf{Comm}_k$ the category of commutative algebras, then the left adjoint to the forgetful functor $\mathbf{Comm}_k \to \mathbf{Vect}_k$ is the functor of the symmetric algebra: $V \mapsto SV$.

One can give many more examples, spanning many fields. These examples show that adjoint functors are ubiquitous in mathematics.

7.7. Abelian categories

The type of categories that most often appears in representation theory is abelian categories. The standard definition of an abelian category is rather long, so we will not give it here, referring the reader to the textbook [**Fr**]; rather, we will use as the definition what is really the statement of the Freyd-Mitchell theorem:

Definition 7.7.1. An **abelian category** is a category (enriched over the category of abelian groups) which is equivalent to a full subcategory $\mathcal{C}$ of the category A-mod of left modules over a ring A, closed under taking finite direct sums, as well as kernels, cokernels, and images of morphisms.